

CASE RECORDS of the MASSACHUSETTS GENERAL HOSPITAL

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Case 17-2026: A 74-Year-Old Man with Pancytopenia

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PRESENTATION OF CASE

Dr. Van-Khue Ton: A 74-year-old man was admitted to this hospital because of persistent pancytopenia that had resulted in multiple red-cell transfusions.

Three months before the current presentation, the patient underwent heart transplantation after a prolonged admission in the cardiac intensive care unit of this hospital. He had been admitted for decompensated heart failure due to transthyretin amyloid cardiomyopathy, for which he received treatment with infusions of inotropic and diuretic agents. The preoperative course was notable for anemia, for which he received intravenous iron therapy. In addition, the patient had persistent thrombocytopenia that prompted a switch in treatment from heparin to a direct thrombin inhibitor, despite the fact that testing for antibodies against platelet factor 4 was negative. Serologic testing performed before transplantation was positive for Epstein–Barr virus (EBV) IgG, negative for cytomegalovirus (CMV) IgG, and negative for toxoplasma IgG. Serologic testing in the donor was positive for EBV IgG and CMV IgG and negative for toxoplasma IgG. Laboratory test results are shown in Table 1.

After transplantation, the patient had intact allograft function, but transient acute kidney injury developed. Milrinone infusion was administered from the time of transplantation until post-transplantation day 8. Endomyocardial biopsies were performed at 7 and 14 days after transplantation, and both specimens showed mild lymphocyte infiltration without myocardial injury. On post-transplantation day 17, the patient was discharged from the hospital. New medications prescribed at discharge included prednisone, mycophenolate mofetil, tacrolimus, valganciclovir, trimethoprim–sulfamethoxazole, and posaconazole. In addition, the patient continued to receive treatment with tafamidis, famotidine, pravastatin, amlodipine, torsemide, calcium, and vitamin D. The results of laboratory tests obtained at the time of discharge are shown in Table 1.

Twenty-eight days after transplantation, the patient returned to the transplantation center at this hospital for outpatient follow-up. Routine post-transplantation right heart catheterization revealed low-normal filling pressures and a cardiac index of 4.1 liters per minute per square meter of body-surface area (reference value, >2.4).

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Table 1. Laboratory Data.*

Variable	Reference Range, Adults†	2 Days before Transplantation, ICU	Day 17 after Transplantation	Day 28 after Transplantation	Day 38 after Transplantation	Day 56 after Transplantation	On Readmission, Day 70 after Transplantation	Day 71 after Transplantation
Hemoglobin (g/dl)	13.5–17.5	13.7	8.8	7.7	6.7	5.6	6.6	7.4
Hematocrit (%)	41.0–53.0	40.5	26.3	22.2	20.1	16.9	18.5	21.7
Reticulocyte count (%)	0.5–2.5	—	—	—	<0.5	<0.5	<0.5	—
Mean corpuscular volume (fl)	80.0–100.0	—	97.0	94.9	94.8	90.9	85.6	88.0
White-cell count (per μ l)	4500–11,000	6190	5640	3800	5290	2950	3690	2560
Differential count (per μ l)								
Neutrophils	1800–7700	—	5200	3310	4700	2320	3080	1410
Lymphocytes	1000–4800	—	270	290	370	310	210	570
Monocytes	200–1200	—	140	140	180	220	290	500
Platelet count (per μ l)	150,000–400,000	89,000	95,000	97,000	94,000	62,000	92,000	57,000
Sodium (mmol/liter)	135–145	136	136	136	140	140	136	139
Potassium (mmol/liter)	3.4–5.0	3.8	4.5	4.5	5.3	5.3	5.3	4.4
Chloride (mmol/liter)	98–108	103	101	100	106	107	104	110
Carbon dioxide (mmol/liter)	23–32	21	23	21	17	18	19	19
Urea nitrogen (mg/dl)	8–25	27	50	83	68	75	67	55
Creatinine (mg/dl)	0.60–1.50	1.14	1.90	2.71	2.20	2.44	2.02	1.80
Cystatin C (mg/liter)	0.61–0.95	—	—	3.26	3.00	2.25	3.19	—
Glucose (mg/dl)	70–110	90	81	132	85	—	109	88
Total protein (g/dl)	6.0–8.3	6.3	5.1	—	6.3	5.6	5.1	5.1
Globulin (g/dl)	1.9–4.1	2.7	1.7	—	—	1.9	1.8	1.5
Albumin (g/dl)	3.3–5.0	3.6	3.4	—	3.8	4.2	4.0	3.6
Alanine aminotransferase (U/liter)	10–55	24	60	—	244	75	27	23
Aspartate aminotransferase (U/liter)	10–40	32	36	—	51	30	27	32
Total bilirubin (mg/dl)	0.0–1.0	1.8	1.5	—	1.0	1.8	1.1	1.7
Direct bilirubin (mg/dl)	0.0–0.4	0.6	0.7	—	—	0.8	0.5	0.7
Lactate dehydrogenase (U/liter)	110–210	—	—	—	250	—	385	—
Direct antibody test	Negative	—	—	—	—	Negative	Negative	—

N-terminal pro-B-type natriuretic peptide (pg/ml)	<900	—	—	—	4515	—	—
Fibrinogen (mg/dl)	150–400	—	—	—	—	2468	—
Haptoglobin (mg/dl)	30–200	—	—	134	—	142	—
25-Hydroxyvitamin D (ng/ml)	33–100	—	—	—	19	—	—
Iron (μg/dl)	30–160	—	—	286	256	297	—
Iron-binding capacity (μg/dl)	230–404	—	—	328	276	318	—
Ferritin (μg/liter)	10–200	—	—	855	1229	1775	—
Vitamin B ₁₂ (pg/ml)	>231	—	—	492	—	592	—
Folate (ng/ml)	>4.7	—	—	6.7	—	5.0	—
Erythropoietin (mIU/ml)	2.6–18.5	—	—	—	—	—	465
Zinc (μg/dl)	60–107	—	—	—	—	—	48
Rheumatoid factor (IU/ml)	0–14	—	—	—	—	—	20
Antinuclear antibody	Negative at 1:40 and 1:160	—	—	—	—	—	Positive at 1:40 and 1:160, homogeneous pattern
IgG (mg/dl)	614–1295	—	—	—	—	—	523
IgA (mg/dl)	69–309	—	—	—	—	—	81
IgM (mg/dl)	53–334	—	—	—	—	—	53
Electrophoresis and immunofixation	No M component	—	—	—	—	—	Normal pattern, mild diffuse decrease in immune globulin
Free kappa light chain (mg/liter)	3.3–19.4	—	—	—	—	—	30.2
Free lambda light chain (mg/liter)	5.7–26.3	—	—	—	—	—	20.9
Kappa:lambda ratio	0.30–1.70	—	—	—	—	—	1.44
Tacrolimus trough (ng/ml)	10–13 during first year after heart transplantation	—	13.2	8.7	13.3	8.0	7.8
Posaconazole (ng/ml)	>1000	—	1290	—	—	—	805

* To convert the values for urea nitrogen to millimoles per liter, multiply by 0.357. To convert the values for creatinine to micromoles per liter, multiply by 88.4. To convert the values for glucose to millimoles per liter, multiply by 0.05551. To convert the values for bilirubin to micromoles per liter, multiply by 17.1. To convert the values for 25-hydroxyvitamin D to nanomoles per liter, multiply by 2.496. To convert the values for iron and iron-binding capacity to micromoles per liter, multiply by 0.1791. To convert the values for vitamin B₁₂ to picomoles per liter, multiply by 0.7378. To convert the values for folate to nanomoles per liter, multiply by 2.266.

† Reference values are affected by many variables, including the patient population and the laboratory methods used. The ranges used at Massachusetts General Hospital are for adults who are not pregnant and do not have medical conditions that could affect the results. They may therefore not be appropriate for all patients.

An endomyocardial biopsy showed no evidence of allograft rejection. Laboratory test results are shown in Table 1. The dose of tacrolimus was decreased to maintain the target serum trough concentration of 10 to 13 ng per milliliter; the prednisone dose was decreased in accordance with institutional protocol.

Thirty-eight days after transplantation, the patient was admitted to the cardiac surgical unit because of a hemoglobin level of 6.7 g per deciliter (reference range, 13.5 to 17.5) and a potassium level of 5.3 mmol per liter (reference range, 3.4 to 5.0). Testing for hepatitis B virus DNA, hepatitis C virus RNA, and human immunodeficiency virus (HIV) RNA was negative. Additional laboratory test results are shown in Table 1.

Dr. Sadiya Sultana: Radiography of the chest (Fig. 1A) showed low lung volumes, along with some bandlike retrocardiac opacities in the left lower lobe, findings that most likely indicated atelectasis. There was no pleural effusion.

Dr. Ton: Two units of packed red cells were transfused, and treatment with oral sodium zirconium cyclosilicate was administered. Genetic testing for the H63D and C282Y variants in the gene *HFE* was negative. Transthoracic echocardiography showed normal biventricular function, dilatation of the right ventricle and left atrium, trace interatrial shunting, and no pericardial effusion. Treatment with trimethoprim-sulfamethoxazole was discontinued, and atovaquone was started for prophylaxis against *Pneumocystis jirovecii* pneumonia and toxoplasmosis. The dose of mycophenolate mofetil was decreased. The potassium level normalized, and on the fourth hospital day, he was discharged home.

Fifty-six days after transplantation, right heart catheterization showed normal filling pressures and a cardiac index of 3.8 liters per minute per square meter of body-surface area. An endomyocardial biopsy was performed, and the specimen showed no evidence of rejection. Later that day in the transplantation clinic, the patient appeared fatigued, and he was admitted to the hospital. Laboratory test results are shown in Table 1. The parathyroid hormone level was normal. A sample of blood was obtained for culture. Two units of packed leukoreduced red cells were transfused, and a third unit was transfused the next day. Testing for CMV DNA was negative. Treatment was switched from valganciclovir to

famciclovir; the doses of mycophenolate mofetil, prednisone, and posaconazole were decreased. The target tacrolimus serum trough level was reduced to limit the risk of nephrotoxicity. On the fifth hospital day, the patient was discharged home.

Nine days after discharge, and 70 days after transplantation, the patient was evaluated at the transplantation center of this hospital. Because of persistent clinically significant anemia, planned right heart catheterization and endomyocardial biopsy were deferred, and the patient was readmitted to the hospital.

A review of systems was notable for persistent fatigue and episodes of epistaxis. The patient reported intermittent tremor in both arms. He had no ecchymoses, petechiae, gum bleeding, hematochezia, or melena. In addition, he had no fever, night sweats, anorexia, cough, gastrointestinal symptoms, or known sick contacts.

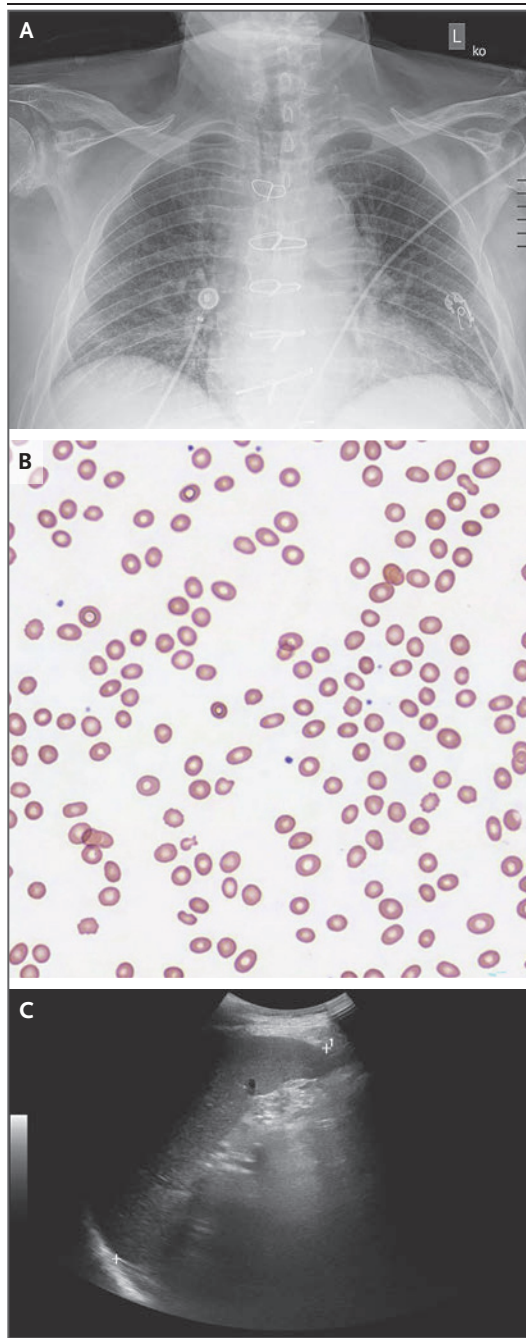
The patient's cardiac history in the native heart included coronary atherosclerosis, transthyretin amyloid cardiomyopathy, heart failure, and atrial dysrhythmia that had been treated with ablation. In addition, his medical history included amyloidosis, hypertension, hyperlipidemia, ischemic stroke that had been treated with carotid thrombectomy and stenting, chronic kidney disease, colonic tubular adenomas, gastroesophageal reflux disease and duodenitis, primary hyperparathyroidism that resulted in parathyroidectomy, carpal tunnel syndrome that had been treated with bilateral carpal tunnel release surgery, pulmonary nodules, and sleep apnea that had been treated with tonsillectomy and uvulopalatopharyngoplasty. Medications included tafamidis, prednisone, mycophenolate mofetil, tacrolimus, atovaquone, famciclovir, famotidine, pravastatin, posaconazole, amlodipine, and torsemide. Dietary supplements included calcium and vitamin D. Treatment with angiotensin-converting-enzyme inhibitors had caused cough.

The patient was a retired consultant and lived in coastal Massachusetts. He smoked marijuana and consumed one beer almost daily. He did not use tobacco or illicit drugs. He was not sexually active. The patient had previously visited California but had not traveled in the past year. His father had had coronary artery disease.

On physical examination, the temporal temperature was 36.3°C, the heart rate 100 beats per

Figure 1. Chest Radiograph, Peripheral-Blood Smear, and Abdominal Ultrasound.

An image obtained on portable chest radiography on post-transplantation day 38 (Panel A) shows low lung volumes with some retrocardiac bandlike opacities, most likely indicating atelectasis, and no evidence of pleural effusion. The patient had previously undergone median sternotomy. A peripheral-blood smear obtained on post-transplantation day 70 (Panel B) shows anisopoikilocytosis, with an absence of reticulocytes. Images from a limited right-upper-quadrant abdominal ultrasound obtained the next day (Panel C) show splenomegaly, with the spleen measuring 16.7 cm in cranio-caudal dimension.



minute, the blood pressure 100/62 mm Hg, and the oxygen saturation 100% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 33.4. He had conjunctival pallor. The sternal surgical incision was well approximated. Auscultation of the heart revealed a regular rhythm without murmurs or a friction rub. He had edema in both legs, which was more pronounced in the right leg than in the left; there was no calf tenderness. Neither ecchymosis nor rash was noted. The remainder of the examination was normal.

The thyrotropin level, prothrombin time, international normalized ratio, and partial thromboplastin time were normal. Other laboratory test results are shown in Table 1.

Dr. Sharl S. Azar: A peripheral-blood smear showed red-cell anisocytosis and central pallor. The white cells appeared normal, and the platelets had normal morphologic features and granularity (Fig. 1B).

Dr. Ton: Electrocardiography was performed and showed sinus tachycardia, intraventricular conduction delay, and baseline artifact. The electrocardiogram also showed nonspecific ST-T segment changes.

Dr. Sultana: In addition, duplex ultrasonography of the right leg showed no deep venous thrombosis. All the visualized veins were fully compressible with normal flow.

Dr. Ton: Two units of packed leukoreduced red cells were transfused. The patient was evaluated by the hematology consultant, and additional testing was performed.

On the second hospital day, the serum ceruloplasmin level was normal. Other laboratory test results are shown in Table 1.

Dr. Sultana: A targeted abdominal ultrasound of the right upper quadrant (Fig. 1C) revealed splenomegaly. The spleen measured 16.7 cm in the cranio-caudal dimension (reference value, <13).

Dr. Ton: Treatment with famciclovir and mycophenolate mofetil was stopped, and treatment with sirolimus (1 mg per day) and folate was started. A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. William C. Aird: This 74-year-old man underwent orthotopic heart transplantation for decompensated heart failure due to transthyretin amyloid cardiomyopathy. In the weeks that followed transplantation, he presented with various combinations of cytopenias, most notably severe transfusion-dependent anemia. Single or multilineage cytopenias occur in 38 to 63% of patients after solid-organ transplantation, usually within the first 100 days after surgery.¹ Decreased cell counts in solid-organ transplant recipients may be classified according to the pathophysiological mechanism, cause, or cell line involved.

PATHOPHYSIOLOGICAL MECHANISMS OF CYTOPENIAS

The pathophysiological mechanisms of cytopenias are not specific to solid-organ transplant recipients and include decreased production, increased destruction, and splenic sequestration. In addition, certain lineage-specific mechanisms should be considered, such as anemia from blood loss, dilutional thrombocytopenia in the context of massive transfusion, and in the case of leukopenia, increased cell trafficking between blood and tissue. Most, but not all, causes of post-transplantation cytopenias stem from decreased production (i.e., myelosuppression).

SPECIFIC CAUSES OF CYTOPENIAS

Specific causes of cytopenias that occur after solid-organ transplantation include medications, infections, inflammation, and immune-mediated processes.¹⁻³ After transplantation, patients may present with a single cytopenia, bicytopenia, or pancytopenia. When multiple cytopenias are present, one cell lineage may be preferentially affected. Moreover, the pattern of cytopenias may change in the same patient over time. In many cases, multiple factors contribute to a decrease in one or more cell lineages.

NARROWING THE FOCUS

I will consider three aspects of this patient's presentation in order to narrow the differential diagnosis. First, the red cells in this patient were preferentially affected. At selected time points after surgery, he had pancytopenia with concomitant decreases in the white-cell count, hemoglobin level, and platelet count. However, the total white-

cell count varied over time, with values ranging from normal to slightly below normal. Moreover, the white-cell differential count revealed persistent lymphocytopenia — a common finding after transplantation. The absolute neutrophil count was normal until the second day of the current admission, when it decreased to just below the lower limit of the normal range. A low lymphocyte count can cause leukopenia; however, the mechanisms, causes, and clinical implications of lymphocytopenia differ from those of neutropenia, although they have overlapping features.⁴⁻⁶ On a practical level, the term pancytopenia is often reserved for cases in which the counts of all three myelopoietic lineages (red cells, platelets, and neutrophils) are reduced, a threshold that was rarely met in this patient.

Second, the timing of the development of the different cytopenias may be an important clue in this patient's case. The absolute lymphocyte count and, more importantly, the platelet count were decreased before surgery. Insufficient information is available regarding the patient's preoperative course to definitively determine the cause of the initial thrombocytopenia. Possible causes include medication effect, infection, implantation of an extracorporeal device, or congestive hypersplenism. Regardless of the cause, thrombocytopenia first manifested before transplantation, whereas severe anemia worsened after transplantation. It seems most likely that these two cytopenias are — at least in part — mechanistically distinct.

Finally, the relative severity of the different cytopenias is instructive in this patient. In contrast to the mild decreases in the white-cell and platelet counts, he had a striking reduction in the hemoglobin level, with a nadir of 5.6 g per deciliter 56 days after transplantation. The severity of the anemia in this patient is so far out of proportion to his leukopenia and thrombocytopenia that it suggests a primary erythroid process.

On the basis of these considerations, I will focus on the causes of post-transplantation anemia.

POST-TRANSPLANTATION ANEMIA

After heart transplantation, the prevalence of anemia ranges from 20 to 51% at different time points, with a prevalence of about 40% at 1 year after the procedure.¹ Anemia may result from blood loss after surgery, immune- or non-immune-

Table 2. Causes of Anemia after Solid-Organ Transplantation.

Hyperproliferative Causes	Hypoproliferative Causes
Bleeding Perioperative bleeding Frequent blood draws	Inflammation Anemia of inflammation Hemophagocytic lymphohistiocytosis
Hemolytic anemia Immune causes ABO-mismatched solid-organ transplantation (passenger lymphocyte syndrome) Drug-induced hemolytic anemia β -lactam antibiotics Trimethoprim–sulfamethoxazole Nonimmune causes Medications associated with thrombotic microangiopathy Cyclosporine Tacrolimus Oxidant medications associated with hemolysis in glucose-6-phosphate dehydrogenase deficiency Dapsone Trimethoprim–sulfamethoxazole Infections Cytomegalovirus infection Chronic hepatitis C virus infection BK polyomavirus infection Babesiosis Malaria	Medications Mycophenolate mofetil Tacrolimus Ganciclovir Valganciclovir Trimethoprim–sulfamethoxazole (may induce megaloblastic anemia) Infections Parvovirus B19 infection Viral hepatitis infection Human immunodeficiency virus infection Epstein–Barr virus infection Cytomegalovirus infection Human herpesvirus-6 infection Human herpesvirus-8 infection Post-transplantation lymphoproliferative disorder

mediated hemolysis, reduced erythropoiesis, or, less commonly, splenic sequestration (Table 2).

When evaluating a patient with anemia, the first branch point is whether the reticulocyte count is inappropriately low (hypoproliferative anemia) or appropriately elevated (hyperproliferative anemia). An appropriately elevated reticulocyte count in the context of anemia almost always indicates bleeding or hemolysis. This reticulocyte count in this patient was profoundly low. Although reticulocytopenia in a patient with anemia indicates a problem with red-cell production or erythropoiesis, it does not rule out concomitant bleeding or hemolysis. However, this patient had no history that would suggest bleeding, and markers of hemolysis — most importantly, the haptoglobin level — are normal. Overall, this patient's presentation is most consistent with red-cell aplasia, a severe type of hypoproliferative anemia.

PURE RED-CELL APLASIA

Pure red-cell aplasia is a syndrome characterized by isolated normocytic normochromic anemia (although macrocytic anemia can also occur), severe reticulocytopenia, and marked reduction or absence of erythroid precursors in the bone marrow. Implicit in the definition is that other causes

of reduced erythropoiesis, including nutritional deficiency, hypothyroidism, and chronic kidney disease, have been ruled out. This patient received intravenous iron preoperatively, so it is unlikely that he would have been iron-deficient after surgery. In addition, the levels of vitamin B₁₂ and thyrotropin were normal, and the creatinine level was only mildly elevated.

During the aplastic phase of pure red-cell aplasia, the levels of serum ferritin and erythropoietin are elevated, with almost complete saturation of iron-binding capacity, as measured by transferrin saturation, which is calculated by multiplying the serum iron level by 100 and dividing the result by the total iron-binding capacity.⁷ The profound anemia causes hypoxia-mediated elevation of erythropoietin. The mechanism of iron overload may involve erythroferrone-mediated suppression of hepcidin expression. In this patient, the ferritin level increased over time, and transferrin saturation was nearly 100%. In addition, the erythropoietin level was markedly elevated on the one occasion that it was measured. Overall, this patient's laboratory test results are suggestive of pure red-cell aplasia.

Pure red-cell aplasia may be congenital or acquired. Acquired pure red-cell aplasia — the most likely diagnosis in this patient — may be

Table 3. Causes of Pure Red-Cell Aplasia.

Congenital
Diamond–Blackfan anemia
Acquired
Primary (i.e., idiopathic)
Secondary
Autoimmune disorders
Systemic lupus erythematosus
Rheumatoid arthritis
General medications
Azathioprine
Isoniazid
Diphenylhydantoin
Recombinant erythropoietin (older preparations)
Transplantation-related medications
Mycophenolate mofetil
Tacrolimus
Trimethoprim–sulfamethoxazole
Lymphoproliferative disorders
Chronic lymphocytic leukemia
Large granular lymphocytic leukemia
Solid tumors
Thymoma
Others
Infections
Parvovirus B19 infection
Human immunodeficiency virus infection
Epstein–Barr virus infection
Viral hepatitis infection
Cytomegalovirus infection
Tuberculosis

primary (i.e., idiopathic) or secondary. Causes of secondary pure red-cell aplasia include autoimmune diseases (e.g., systemic lupus erythematosus or rheumatoid arthritis), lymphoproliferative disorders (particularly chronic lymphocytic leukemia and T-cell large granular lymphocytic leukemia), certain solid tumors (most prominently thymoma), infections, and medications (Table 3).⁸ This patient did not have a history of autoimmune disease. The peripheral-blood smear showed normal white-cell morphologic features without evidence of an abnormal lymphoid population, and the lactate dehydrogenase level was normal, arguing against the presence of a lymphoproliferative disorder. Although it is a consideration in this patient, a post-transplantation lymphoproliferative disorder typically manifests with multiple cell lines involved as opposed to pure red-cell aplasia. Thymoma was not suspected in this patient. Infection with HIV and CMV was ruled out. The remaining possible causes include other infections (e.g., parvovirus B19 or EBV) or medications (e.g., mycophenolate mofetil,⁹ tacrolimus,¹⁰ or rimethoprim–sulfamethoxazole¹¹). Of all the infectious causes,

parvovirus B19 infection is the most common cause of pure red-cell aplasia.

PARVOVIRUS B19 INFECTION

Parvovirus B19 is a single-stranded DNA virus that preferentially infects erythroblasts.¹² It enters the erythroid progenitor cell through the erythrocyte P-antigen receptor. Viral replication produces a distinctive cytopathic effect, which can be seen with the use of light microscopy in the bone marrow and is characterized by giant proerythroblasts. Human parvovirus B19 is ubiquitous; it infects approximately half the population by 15 years of age.¹³

In immunocompetent hosts, parvovirus B19 infection suppresses erythropoiesis for approximately 1 week, which leads to a reduced reticulocyte count and a decrease in hemoglobin to low-normal or slightly low levels.¹² Viremia is followed by the appearance of virus-specific IgM antibodies within 10 to 12 days after infection, and then IgG antibodies at 2 weeks.¹² In adults, parvovirus B19 infection can be asymptomatic or symptomatic, with fever and malaise followed by arthralgia and rash.

In immunocompromised patients, including solid-organ transplant recipients, a robust humoral immune response to parvovirus B19 is unlikely. Therefore, parvovirus B19 infection in these patients often leads to persistent viremia, indefinite suppression of erythropoiesis, and the development of chronic severe anemia or pure red-cell aplasia.¹²

In published studies, estimates of the incidence of parvovirus B19 infection among solid-organ transplant recipients vary widely, which makes the true risk difficult to determine. Among seronegative solid-organ transplant recipients, the estimated incidence rate is 1.2 cases per 100 recipients during the first month after transplantation and 0.21 cases per 100 recipients per month from the second month through the sixth month.¹⁴

The clinical manifestations of parvovirus B19 infection in patients who have undergone solid-organ transplantation may be atypical. Fever, arthralgia, and rash are present in a minority of these patients. Anemia is a universal feature that is characterized not only by profound reticulocytopenia but also by a lack of response to erythropoietin.

In one review, 16 cases of parvovirus B19–induced severe anemia in heart transplant recipients were reported across a broad age range.¹⁵ The mean hemoglobin level before diagnosis was 5.9 g per deciliter, and most of the cases were transfusion dependent. The authors concluded that parvovirus B19 infection should be considered in heart transplant recipients who present with severe anemia of unknown cause.¹⁵

Occasionally, parvovirus B19 infection can manifest with pancytopenia in patients after solid-organ transplantation. As much as I have downplayed the pancytopenia in this patient and focused on anemia instead of the other cell lines, I acknowledge that the laboratory test results on the current admission point to an evolving panmyelopoietic process, with not only a reduction in the hemoglobin level, white-cell count, and platelet count, but also new neutropenia and worsening of thrombocytopenia.

The notion that parvovirus B19 infection can cause pancytopenia in patients who undergo transplantation is well established. The American Society of Transplantation Infectious Diseases Community of Practice 2019 guideline on human parvovirus B19 strongly recommends that parvovirus B19 infection be suspected in solid-organ transplant recipients with pancytopenia.¹⁶

HEMOPHAGOCYTIC LYMPHOHISTIOCYTOSIS

The development of pancytopenia and elevated levels of ferritin in a patient who has undergone transplantation increases the possibility of hemophagocytic lymphohistiocytosis (HLH).¹⁷⁻¹⁹ Previous studies showed that 40% of HLH cases associated with solid-organ transplantation occurred within the first 3 months after transplantation, and approximately 67% occurred within the first year.¹⁷ In many cases, infections, including parvovirus B19 infection,^{17,20} are identified as possible triggers. There is no single clinical or laboratory feature with sufficient sensitivity and specificity to definitively diagnose HLH. Clinical scoring systems have not been validated in patients with HLH associated with solid-organ transplantation. This patient had nonspecific findings of cytopenias, elevated levels of aspartate aminotransferase and alanine aminotransferase, and splenomegaly. However, other HLH markers were either negative or not reported. Moreover, the elevated ferritin level was associated with increased trans-

ferrin saturation, a finding that suggests an iron overload state rather than an inflammatory cause of hyperferritinemia. Overall, my suspicion of HLH is low.

SUMMARY

I suspect this patient has pure red-cell aplasia due to parvovirus B19 infection. I would obtain serologic testing (IgG and IgM) and quantitative nucleic acid testing of serum for parvovirus B19.¹⁶ Serologic testing may be less helpful than nucleic acid testing in solid-organ transplant recipients because an antibody response is often absent.¹⁴ A bone marrow biopsy should be performed if serologic and nucleic acid tests in solid-organ transplant recipients are negative.

CLINICAL IMPRESSION

Dr. Azar: Initially, this patient's cytopenias were thought to be the result of the immunosuppressive medication regimen. However, the cytopenias continued to worsen despite adjustments to these medications. While other causes of cytopenias were being considered, a bone marrow biopsy was performed to rule out an intrinsic bone marrow process and to facilitate safe dose adjustments of immunosuppressive medications moving forward.

CLINICAL DIAGNOSIS

Intrinsic process of the bone marrow causing cytopenias.

DR. WILLIAM C. AIRD'S DIAGNOSIS

Pure red-cell aplasia due to parvovirus B19 infection.

DIAGNOSTIC TESTING

Dr. Lisa D. Yuen: The bone marrow aspirate smear (Fig. 2A) showed trilineage hematopoiesis, with megakaryocytes, myeloid elements, and erythroid elements arrested at the proerythroblast stage. There were numerous large proerythroblasts, with deeply basophilic cytoplasm, round nuclei, fine chromatin, and eosinophilic nuclear inclusions, findings suggestive of infection by parvovirus B19. Erythroids at more-differentiated precursor stages were absent. Examination of a core-biopsy

specimen of bone marrow (Fig. 2B) showed a normocellular marrow with complete myeloid maturation to segmented neutrophils. Megakaryocytes were adequate in number and had morphologically normal forms. Erythroid maturation was markedly left-shifted, with a paucity of mature forms. Numerous giant proerythroblasts with well-demarcated eosinophilic intranuclear inclusions were present. Although such inclusions are rare in immunocompetent patients with parvovirus B19 infection, they are abundant in immunosuppressed adults and fetuses because of a lack of effective IgG antibody production, which leads to persistent infection.²¹

Dr. Azar: The findings suggestive of parvovirus B19 infection from the bone marrow biopsy were unexpected; however, in hindsight, they accounted for this patient's presentation. Serologic and nucleic acid testing for parvovirus B19 was performed.

Dr. Yuen: Serologic testing for antibodies against parvovirus B19 detected IgM antibodies but not IgG antibodies. Quantitative nucleic acid testing was also performed to measure the amount of parvovirus B19 DNA in the patient's blood, which exceeded the upper limit of the assay (>100,000,000 DNA copies per milliliter).

Immunohistochemical staining was performed

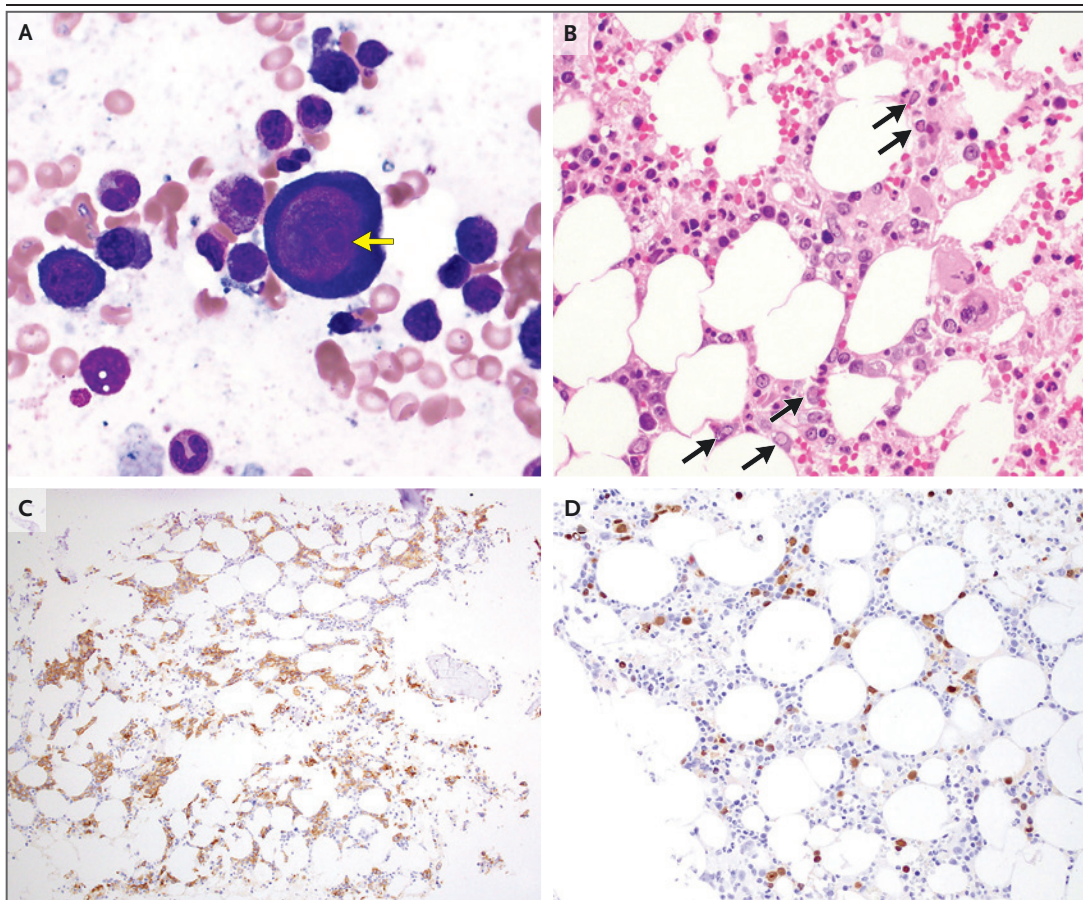


Figure 2. Bone Marrow Aspirate and Core-Biopsy Specimen.

Wright–Giemsa staining of the smear of bone marrow aspirate (Panel A) shows a giant proerythroblast with an eosinophilic intranuclear inclusion (arrow), a finding that indicates parvovirus B19 infection. Hematoxylin and eosin staining of the bone marrow core-biopsy specimen (Panel B) reveals trilineage hematopoiesis with maturing myeloid elements, megakaryocytes, and marked left-shifted erythroid elements, which includes many giant proerythroblasts with eosinophilic intranuclear inclusions (arrows). Immunohistochemical staining for E-cadherin performed on the core-biopsy specimen (Panel C) highlights erythroid progenitors, which are increased in number and show frequent clustering. Immunohistochemical staining for parvovirus B19 (Panel D) is strongly positive in many of the erythroid precursors.

on the biopsy specimen. Immunohistochemical staining for E-cadherin highlighted erythroid precursors, which were increased in number and showed frequent clustering (Fig. 2C). Parvovirus B19–specific immunostaining was strongly positive in many erythroid precursors (Fig. 2D), which confirmed that the patient's severe, persistent anemia was caused by parvovirus B19 infection. Because of the history of chronic pancytopenia and presence of some small megakaryocytes with hypolobated nuclei on the bone marrow biopsy, the possibility of myelodysplastic syndrome was considered. However, cytogenetic analysis of the bone marrow aspirate showed a normal karyotype, and DNA sequencing detected no pathogenic mutations. Therefore, no evidence of a hematopoietic neoplasm was present.

PATHOLOGICAL DIAGNOSIS

Parvovirus B19 infection.

DISCUSSION OF MANAGEMENT

Dr. Camille N. Kotton: There are no direct-acting antiviral agents for parvovirus B19. Treatment in immunocompromised organ-transplant recipients primarily consists of careful dose reduction of transplantation-related immunosuppressive medications, intravenous immune globulin (IVIG) therapy, and supportive care with red-cell transfusions; these were the mainstays of this patient's treatment.

After five doses of IVIG, the parvovirus B19 viral load decreased from 100 million DNA copies per milliliter to 665,200 DNA copies per milliliter. If there had been no response to the first course of IVIG, additional courses of IVIG would

have been considered. Routine preemptive use of additional courses of IVIG to prevent recurrences is not recommended.¹⁶

Parvovirus B19 infection can recur in immunocompromised patients; in one case series, 25% of organ-transplant recipients had a recurrence.²² Given the risk of recurrence, we monitored quantitative parvovirus B19 DNA levels in this patient, along with the complete blood count, including the reticulocyte count. Several months after the patient's initial treatment, he had a recurrent infection. He received treatment with an additional course of IVIG, which resulted in a return of the reticulocyte count to a normal level and a decrease in his quantitative parvovirus B19 DNA level.

Dr. Ton: Seven months after the patient underwent orthotopic heart transplantation, treatment with sirolimus was stopped in order to further reduce immunosuppression. The patient continues to receive tacrolimus with a lower target trough level and a low dose of prednisone. To date, no allograft rejection has occurred, and cardiac function has remained normal.

FINAL DIAGNOSIS

Parvovirus B19 infection.

This case was presented at Medicine Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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